Yarim va chorak doira yuzi

1-test (oson) · 20 ta savol

Ism-familiya: _____ Sinf: _____ Sana: _____

Har savolda bitta to'g'ri javobni belgilang.

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1. **Yarim doira yuzi qaysi formula bilan topiladi?**

A) $\frac{1}{2} \cdot \pi r^2$ **B)** $\frac{1}{2} \cdot 2\pi r$ **C)** $\frac{1}{4} \cdot \pi r^2$ **D)** πr^2

2. **Chorak doiraning yuzi qaysi formula bilan topiladi?**

A) $\frac{1}{4} \cdot \pi r^2$ **B)** $\frac{1}{2} \cdot \pi r^2$ **C)** $4\pi r^2$ **D)** $\frac{1}{4} \cdot 2\pi r$

3. **Doiraning yuzi 50 cm². Shu doiraning yarmidan iborat yarim doiraning yuzi qancha?**

A) 100 cm² **B)** 25 cm² **C)** 12,5 cm² **D)** 50 cm²

4. **Doiraning yuzi 36 m². Shu doiraning choragi yuzi qancha?**

A) 9 m² **B)** 18 m² **C)** 144 m² **D)** 12 m²

5. **Yarim doiraning radiusi 2 cm. Uning yuzini toping (π = 3,14).**

A) 3,14 cm² **B)** 6,28 cm² **C)** 12,56 cm² **D)** 25,12 cm²

6. **Yarim doiraning diametri 10 cm. Uning yuzini toping (π = 3,14).**

A) 157 cm² **B)** 39,25 cm² **C)** 15,7 cm² **D)** 78,5 cm²

7. **Chorak doiraning radiusi 1 m. Uning yuzini toping (π = 3,14).**

A) 0,25 m² **B)** 3,14 m² **C)** 0,785 m² **D)** 1,57 m²

8. **Chorak doiraning radiusi 2,8 cm. Uning yuzini toping (π = $\frac{22}{7}$).**

A) 6,16 cm² **B)** 24,64 cm² **C)** 4,4 cm² **D)** 12,32 cm²

9. **Yarim doiraning diametri 2,8 m. Uning yuzini toping (π = $\frac{22}{7}$).**

A) 1,54 m² **B)** 6,16 m² **C)** 3,08 m² **D)** 12,32 m²

10. **Radiuslari teng bo'lgan ikkita yarim doira birga qanday shakl hosil qiladi?**

A) yarim doira **B)** chorak doira
C) halqa **D)** butun doira

11. **Chorak doiraning yuzi 7 cm². Shunday 4 ta bir xil chorak doira birga butun doira hosil qildi. Doiraning yuzi qancha?**

A) 28 cm² **B)** 11 cm² **C)** 7 cm² **D)** 14 cm²

12. **Dumaloq pitsaning to'rtdan bir bo'lagi yeyildi. Qolgan pitsaning yuzi butun pitsa yuzining qanday qismiga teng?**

A) $\frac{1}{4}$ **B)** $\frac{3}{4}$ **C)** $\frac{1}{2}$ **D)** $\frac{2}{3}$

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13. **Rasmdagi yarim doiraning yuzini toping (π = 3,14).**



A) 100,48 m² **B)** 200,96 m²

C) 50,24 m² **D)** 401,92 m²

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14. **Rasmdagi chorak doiraning yuzini toping (π = 3,14).**



A) 18,84 cm² **B)** 226,08 cm²

C) 113,04 cm² **D)** 452,16 cm²

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15. **Yarim doiraning radiusi santimetrda berilgan. Uning yuzi qaysi birlikda chiqadi?**

A) cm **B)** m **C)** cm³ **D)** cm²

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16. **Ali radiusi 6 cm bo'lgan yarim doiraning yuzini $\frac{1}{2} \cdot 3{,}14 \cdot 6 \cdot 2$ deb hisobladi. Uning xatosi nimada?**

A) diametrni qo'shmagan **B)** π noto'g'ri olingan

C) r^2 o'rniga $r \cdot 2$ ni olgan **D)** $\frac{1}{2}$ ni olmagan

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17. **Yarim doira yuzi $\frac{1}{2} \cdot 3{,}14 \cdot 5 \cdot 5 = 39{,}25$ cm² kabi topildi. Bu yerda 5 soni nima?**

A) radius **B)** diametr

C) perimetr **D)** yarim doira yuzi

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18. **Yarim doiraning yuzini topishda diametr qo'shiladimi?**

A) Ha, radius ham qo'shiladi

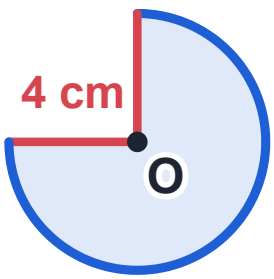
B) Ha, har doim qo'shiladi

C) Ha, faqat π = 3,14 bo'lsa

D) Yo'q, diametr perimetrga qo'shiladi

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19. **Rasmda doiraning $\frac{3}{4}$ qismi, O — markaz. Shaklning yuzini toping (π = 3,14).**



A) 37,68 cm² **B)** 12,56 cm²

C) 50,24 cm² **D)** 25,12 cm²

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20. Yarim doiraning yuzi 20 cm². Radiusi xuddi shunday bo'lgan butun doiraning yuzi qancha?

A) 80 cm² **B)** 40 cm² **C)** 20 cm² **D)** 10 cm²

Javoblar

Yarim va chorak doira yuzi · 1-test (oson)

№	Javob	Qiymati
1	**A**	$\frac{1}{2}$ · πr²
2	**A**	$\frac{1}{4}$ · πr²
3	**B**	25 cm²
4	**A**	9 m²
5	**B**	6,28 cm²
6	**B**	39,25 cm²
7	**C**	0,785 m²
8	**A**	6,16 cm²
9	**C**	3,08 m²
10	**D**	butun doira

№	Javob	Qiymati
11	**A**	28 cm²
12	**B**	$\frac{3}{4}$
13	**A**	100,48 m²
14	**C**	113,04 cm²
15	**D**	cm²
16	**C**	r² o'rniga r · 2 ni olgan
17	**A**	radius
18	**D**	Yo'q, diametr perimetrga qo'shiladi
19	**A**	37,68 cm²
20	**B**	40 cm²

Qisqa yechimlar

1. Yarim doira — doiraning yarmi. Doira yuzi πr², uning yarmi: $\frac{1}{2}$ **· πr²**.

2. Chorak doira — doiraning to'rtdan bir qismi.
Yuzi: $\frac{1}{4}$ **· πr²**. ($\frac{1}{4}$ · 2πr — chorak yoyning uzunligi.)

3. Yarim doira yuzi — doira yuzining yarmi.
50 : 2 = **25 cm²**.

4. Chorak — to'rtdan bir qism.
36 : 4 = **9 m²**.

5. r² = 2 · 2 = 4.
$\frac{1}{2}$ · 3,14 · 4 = **6,28 cm²**.
(12,56 — ikkiga bo'linmagan, bu butun doira.)

6. Radius: 10 : 2 = 5 cm.
$\frac{1}{2}$ · 3,14 · 5 · 5 = **39,25 cm²**.
(157 — radius o'rniga diametr olingan.)

7. $\frac{1}{4}$ · 3,14 · 1 · 1 = 3,14 : 4 = **0,785 m²**.

8. $\frac{1}{4}$ · $\frac{22}{7}$ · 2,8 · 2,8. Avval 2,8 : 7 = 0,4.
$\frac{1}{4}$ · 22 · 0,4 · 2,8 = **6,16 cm²**.
(4,4 cm — bu chorak yoyning uzunligi, yuz emas.)

9. Radius: 2,8 : 2 = 1,4 m. Avval 1,4 : 7 = 0,2.
$\frac{1}{2}$ · 22 · 0,2 · 1,4 = **3,08 m²**.

10. Yarim + yarim = bitta butun. Ikkita bir xil yarim doira birga **butun doira** hosil qiladi.

11. Butun doira 4 ta chorakdan iborat.
4 · 7 = **28 cm²**.

12. Butun pitsa — $\frac{4}{4}$. Choragi yeyildi: $\frac{4}{4} - \frac{1}{4} = \frac{3}{4}$.

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13. Diametr 16 m, radius: 16 : 2 = 8 m.

$\frac{1}{2}$ · 3,14 · 8 · 8 = **100,48 m²**.

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14. $\frac{1}{4}$ · 3,14 · 12 · 12 = 3,14 · 36 = **113,04 cm²**.

(18,84 cm — chorak yoyning uzunligi.)

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15. Yuz kvadrat birliklarda o'lchanadi: **cm²**.

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16. r² = r · r = 6 · 6, 6 · 2 emas.

To'g'risi: $\frac{1}{2}$ · 3,14 · 6 · 6 = 56,52 cm².

Xato: **r² o'rniga r · 2 ni olgan**.

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17. Formulada $\frac{1}{2}$ · πr²: r — **radius**, u ikki marta ko'paytiriladi.

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18. Yuz — shakl ichidagi joy, unga chiziq uzunligi qo'shilmaydi.

Javob: **yo'q, diametr perimetrga qo'shiladi**.

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19. Doira: 3,14 · 4 · 4 = 50,24 cm².

$\frac{3}{4}$ qismi: 50,24 : 4 · 3 = **37,68 cm²**.

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20. Butun doira — ikkita yarim doira.

2 · 20 = **40 cm²**.
